

LAB: 04

This lab focuses on recovering digital photograph evidence.

Objective:

- Learn forensic recovery utilities

Learning Activities:

At the end of these activities, you should understand:

- Understand the file headers
- How to locate altered file header
- How to recover graphic files

Tools to use:

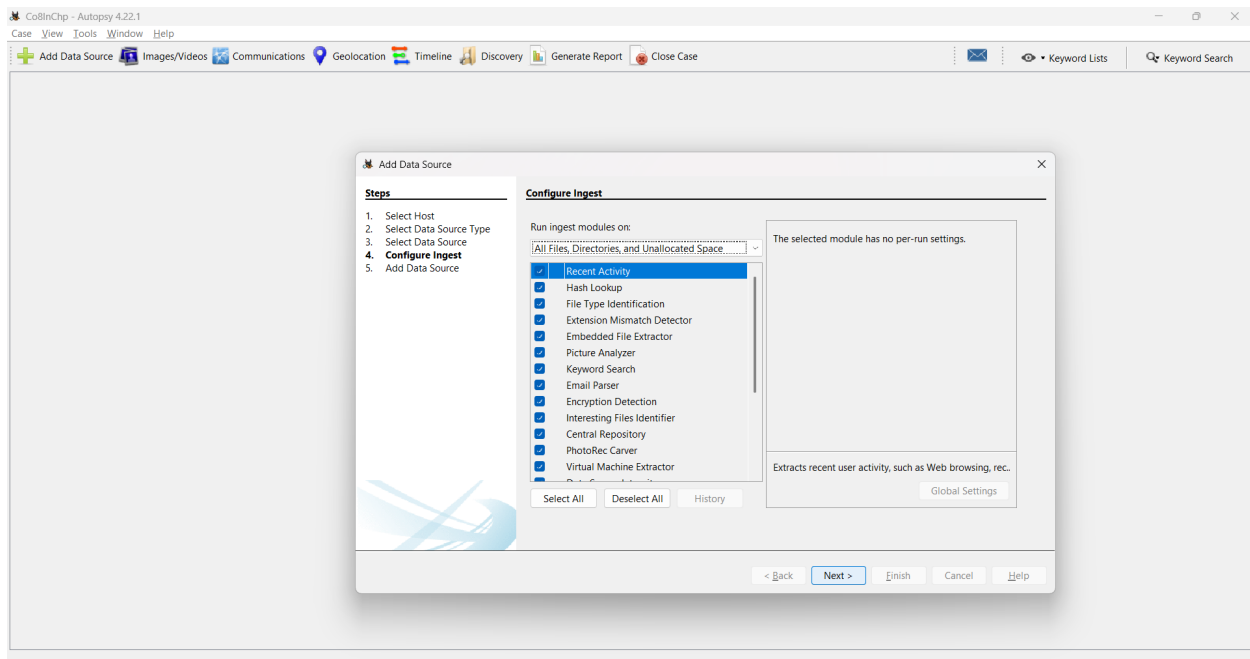
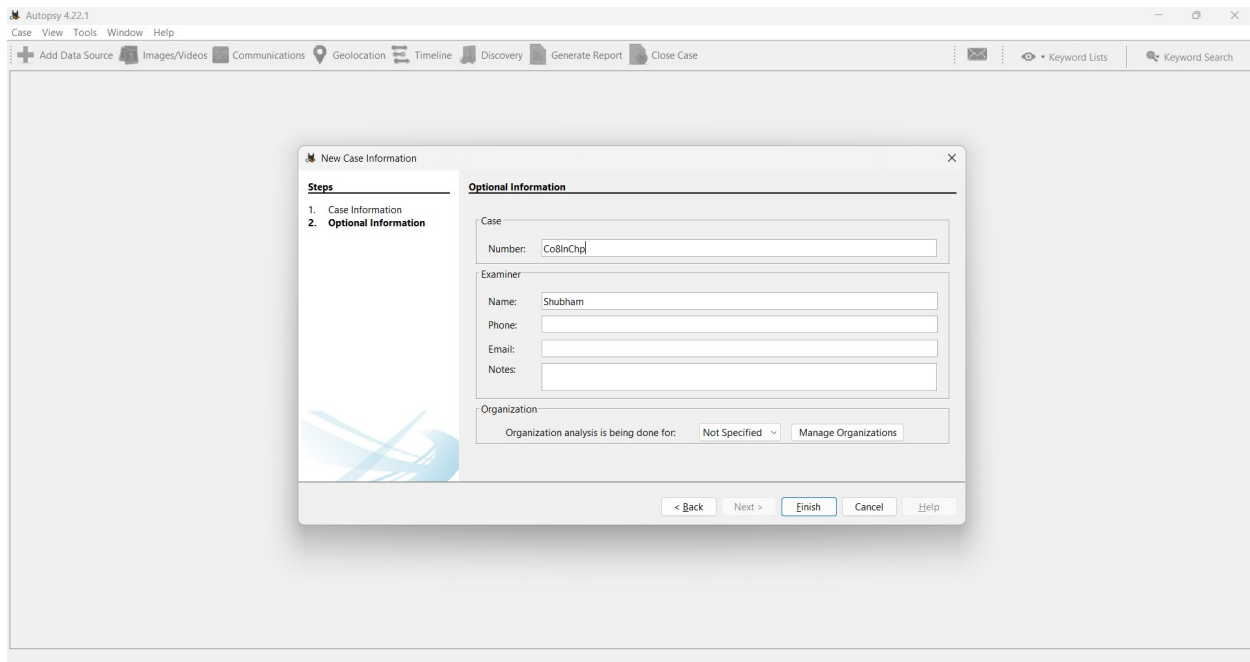
- C08InChp.dd
- Autopsy
- Winhex
- HxD Hex Editor (<https://mh-nexus.de/en/downloads.php?product=HxD20>)

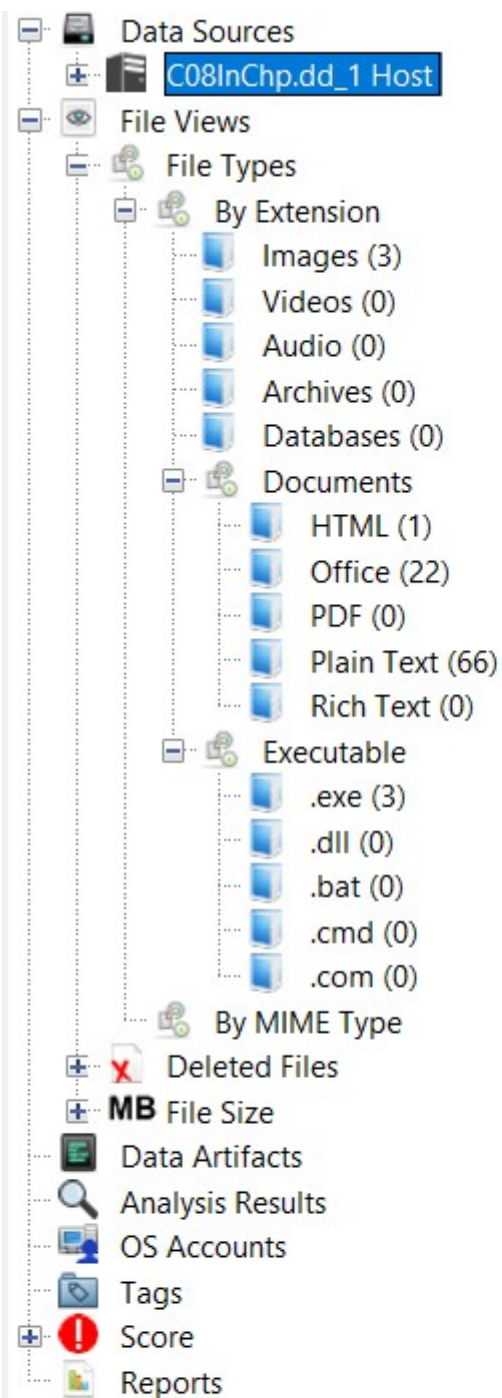
Note: Please include a screenshot of each step. Crop your screenshot to show relevant information only. Uncropped screenshots will result in a 10% deduction from your marks.

Task 1:

Text book, Page 352-355 (Steps 1-9)

Page 357-358 Steps (1-3)





The screenshot shows the Autopsy 4.22.1 interface. The left sidebar has a red box around the 'Analysis Results' section, specifically highlighting 'EXIF Metadata (5)'. The main window displays a table of EXIF metadata for the file '_MG_1345.jpg'.

Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Date Created	Device Model	Device Make
_MG_1345.jpg				File	Not Notable				2009-04-05 18:39:04 EDT	Canon PowerShot SD870 IS	Canon
_R2WHGRN.txt				File	Not Notable				2001-08-07 11:50:49 EDT	Dimage 2330 Zoom	Minolta Co., Ltd
_RBATNM9.jpg				File	Not Notable				2009-04-05 18:39:04 EDT	Canon PowerShot SD870 IS	Canon
f0006352.jpg			0	File	Not Notable				2009-04-05 18:39:04 EDT	Canon PowerShot SD870 IS	Canon

Substring keyword search of “fif”:

Keyword search 33 Results

Table Thumbnail Summary

Save Table as CSV

Name	Keyword Preview	Location	Modified Time	Change Time
_RANH86F.txt	at once, there are «fifty»-two chosen youths f	/img_C08lnChp.dd/\$RECYCLE.BIN/_RANH86F.txt	2007-02-01 13:11:50 EST	0000-00-00 00:00:00
Unalloc_4_1536000_10485760	le work, and on the «fifth» calypso sent him fr	/img_C08lnChp.dd/\$Unalloc/Unalloc_4_1536000_10485760	0000-00-00 00:00:00	0000-00-00 00:00:00
_RBATNM9.jpg	«jfif»"pexifcanoncanon p	/img_C08lnChp.dd/\$RECYCLE.BIN/_RBATNM9.jpg	2017-07-10 17:50:56 EDT	0000-00-00 00:00:00
_5.XLS	ocks of sheep, with «fifty» head in each flock.	/img_C08lnChp.dd/Accounts/GovernmentDATA/_5.XLS	2007-02-01 13:33:28 EST	0000-00-00 00:00:00
f0000192.txt	le work, and on the «fifth» calypso sent him fr	/img_C08lnChp.dd/\$CarvedFiles/1/f0000192.txt	0000-00-00 00:00:00	0000-00-00 00:00:00
_51.XLS	zzzz«zfif»exif minol	/img_C08lnChp.dd/Accounts/GovernmentDATA/_51.XLS	2007-02-01 13:34:06 EST	0000-00-00 00:00:00
f0000280.txt	at table, there are «fifty» maid servants in th	/img_C08lnChp.dd/\$CarvedFiles/1/f0000280.txt	0000-00-00 00:00:00	0000-00-00 00:00:00
f0000784.txt	at once, there are «fifty»-two chosen youths f	/img_C08lnChp.dd/\$CarvedFiles/1/f0000784.txt	0000-00-00 00:00:00	0000-00-00 00:00:00
f0001008.txt	n though there were «fiftv» bands of men surrou	/img_C08lnChp.dd/\$CarvedFiles/1/f0001008.txt	0000-00-00 00:00:00	0000-00-00 00:00:00

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of 1 Page Matches on page: 1 of 1 Match 100% Reset Text Source: Search Results

n cannot stand against many and brave ones. There are not ten suitors only, nor twice ten, but ten many times over; you shall learn their number at once. There are **fifty**-two chosen youths from Duli chium, and they have six servants; from Same there are twenty-four; twenty young Achaeans from Zacynthus, and twelve from Ithaca itself, all of them well born. They have with them a servant Medon, a bard, and two men who can carve at table. If we face such numbers as this, you may have bitter cause to rue your coming, and your revenge. See whether you cannot think of some one who would be willing to come and help us."

"Listen to me," replied Ulysses, "and think whether Minerva and her father Jove may seem sufficient, or whether I am to try and find some one else as well."

"Those whom you have named," answered Telemachus, "are a couple of good allies, for though they dwell high up among the clouds they have power over both gods and men."

"These two," continued Ulysses, "will not keep long out of the fray, when the suitors and we join fight in my house. Now, therefore, return home early to-morrow morning, and go about among the suitors as before. Later on the swineherd will bring me to the city disguised as a miserable old beggar. If you see them ill-treating me, steel your heart against my sufferings; even though they drag me feet foremost out of the house, or throw things at me, look on and do nothing beyond gently trying to make them behave more reasonably; but they will not listen to you, for the day of their reckoning is at hand. Furthermore I say, and lay my saying to your heart, when Minerva shall put it in my mind, I will nod my head to you, and on seeing me do this you must collect all the armour that is in the house and hide it in the strong store room. Make some excuse when the suitors ask you why you are removing it; say that you have taken it to be out of the way of the smoke, inasmuch as it is no longer what it was when Ulysses went away, but has become soiled and begrimed with soot. Add to this more particularly that you are afraid Jove may set them on to quarrel over their wine, and that they may do each other some harm which may disgrace both banquet and wooing, for the sight of arms sometimes tempts people to use them. But leave a sword and a spear apiece for yourself and me, and a couple oxbow shields so that we can snatch them up at any moment; Jove and Minerva will then soon quiet these people. There is also another matter; if you are indeed my son and my blood runs in your veins, let no one know that I Ulysses is within the house; neither Laertes, nor yet the swineherd, nor any of the servants, nor even Penelope herself. Let you and me enjoy

“zzzz” found in file named “gametour2.exe”. After looking at the hex values we can clearly see that the file was tampered with because it has “zzzz” followed by “zFIF” which shows that the file was “JFIF” but the change was sloppy and left the traces of changing the file type, also executable files does not start with “zzzz” rather it starts with “MZ” and have hex value starting with “4D 5A”:

Keyword search 33 Results

Table Thumbnail Summary

Save Table as CSV

Name	Keyword Preview	Location	Modified Time	Change Time
Odessey05.txt	le work, and on the «fifth» calypso sent him fr	/img_C08InChp.dd/Homework/Odessey05.txt	2007-02-01 13:03:44 EST	0000-00-00 00:00:00
_R63H7RL.txt	n though there were «fifty» bands of men surrou	/img_C08InChp.dd/\$RECYCLE.BIN/_R63H7RL.txt	2007-02-01 13:13:54 EST	0000-00-00 00:00:00
Odessey07.txt	at table. there are «fifty» maid servants in th	/img_C08InChp.dd/Homework/Odessey07.txt	2007-02-01 13:04:44 EST	0000-00-00 00:00:00
gametour2.exe	zzzz«zfif«exif minol	/img_C08InChp.dd/Vacation Pictures/gametour2.exe	2001-08-05 07:50:24 EDT	0000-00-00 00:00:00
gametour3.exe	zzzz«zfif«exif minol	/img_C08InChp.dd/Vacation Pictures/gametour3.exe	2001-08-07 04:51:44 EDT	0000-00-00 00:00:00
_REVVHBI.txt	le work, and on the «fifth» calypso sent him fr	/img_C08InChp.dd/\$RECYCLE.BIN/_REVVHBI.txt	2007-02-01 13:03:44 EST	0000-00-00 00:00:00
gametour4.exe	zzzz«zfif«exif minol	/img_C08InChp.dd/Vacation Pictures/gametour4.exe	2001-08-08 08:23:54 EDT	0000-00-00 00:00:00
Odessey12.txt	ocks of sheep, with «fifty» head in each flock.	/img_C08InChp.dd/Homework/Odessey12.txt	2007-02-01 13:09:18 EST	0000-00-00 00:00:00
NTORV_4.TXT	three hundred and fifty years before	/img_C08InChp.dd/\$O-herFiles/NTORV_4.TXT	2007-02-01 13:03:54 EST	0000-00-00 00:00:00

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

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0x00000000: 7A 7A 7A 7A 00 10 7A 46 49 46 00 01 01 01 00 78 zzzz«zfif....x
0x00000010: 00 78 00 00 FF E1 03 1C 45 78 69 66 00 00 49 49 .x.....Exif...II
0x00000020: 2A 00 08 00 00 00 0B 00 0E 01 02 00 0A 00 00 00 ".....
0x00000030: 92 00 00 00 0F 01 02 00 12 00 00 00 9C 00 00 00 .....
0x00000040: 10 01 02 00 12 00 00 00 AE 00 00 00 12 01 03 00 .....
0x00000050: 01 00 00 00 01 00 08 00 1A 01 05 00 01 00 00 00 .....
0x00000060: C0 00 00 00 1B 01 05 00 01 00 00 00 C8 00 00 00 .....
0x00000070: 28 01 03 00 01 00 00 00 02 00 97 02 31 01 02 00 (. .....1...
0x00000080: 0A 00 00 00 D0 00 00 00 32 01 02 00 14 00 00 00 .....2.....
0x00000090: DA 00 00 00 13 02 03 00 01 00 00 00 02 00 97 02 .....
0x000000a0: 69 87 04 00 01 00 00 00 EE 00 00 00 00 00 00 00 i.....
0x000000b0: 20 20 20 20 20 20 20 20 20 00 4D 69 6E 6F 6C 74 .Minolt
0x000000c0: 61 20 43 6F 2E 2C 20 4C 74 64 20 00 44 69 6D 61 a Co., Ltd .Dima
0x000000d0: 67 65 20 32 33 33 30 20 5A 6F 6F 6D 20 00 48 00 ge 2330 Zoom .H.
0x000000e0: 00 00 01 00 00 00 48 00 00 00 01 00 00 00 20 20 .....H.....
0x000000f0: 20 20 20 20 20 20 20 20 32 30 30 31 3A 30 38 3A .....2001:08:
0x00000100: 30 35 20 31 34 3A 35 30 3A 30 37 00 10 00 27 88 05 14:50:07...'
0x00000110: 03 00 04 00 00 00 B4 01 00 00 00 90 07 00 04 00 .....
0x00000120: 00 00 30 32 31 30 03 90 02 00 14 00 00 00 BC 01 ..0210.....

```

File metadata of “gametour2.exe” which also shows that the file is not an executable but an image file because only image file contains such data in the file i.e. file is from a camera or phone and not an “.exe” file:

Keyword search 33 Results

Table Thumbnail Summary

Save Table as CSV

Name	Keyword Preview	Location	Modified Time	Change Time
Odessey05.txt	le work, and on the «fifth» calypso sent him fr	/img_C08InChp.dd/Homework/Odessey05.txt	2007-02-01 13:03:44 EST	0000-00-00 00:00:00
_R63H7RL.txt	n though there were «fifty» bands of men surro	/img_C08InChp.dd/\$RECYCLE.BIN/_R63H7RL.txt	2007-02-01 13:13:54 EST	0000-00-00 00:00:00
Odessey07.txt	at table. there are «fifty» maid servants in th	/img_C08InChp.dd/Homework/Odessey07.txt	2007-02-01 13:04:44 EST	0000-00-00 00:00:00
gametour2.exe	zzzz«zfif»exif minol	/img_C08InChp.dd/Vacation Pictures/gametour2.exe	2001-08-05 07:50:24 EDT	0000-00-00 00:00:00
gametour3.exe	zzzz«zfif»exif minol	/img_C08InChp.dd/Vacation Pictures/gametour3.exe	2001-08-07 04:51:44 EDT	0000-00-00 00:00:00
_REVVHBI.txt	le work, and on the «fifth» calypso sent him fr	/img_C08InChp.dd/\$RECYCLE.BIN/_REVVHBI.txt	2007-02-01 13:03:44 EST	0000-00-00 00:00:00
gametour4.exe	zzzz«zfif»exif minol	/img_C08InChp.dd/Vacation Pictures/gametour4.exe	2001-08-08 08:23:54 EDT	0000-00-00 00:00:00
Odessey12.txt	ocks of sheep, with «fifty» head in each flock.	/img_C08InChp.dd/Homework/Odessey12.txt	2007-02-01 13:09:18 EST	0000-00-00 00:00:00
NTONY_1.TXT	three hundred and fifty years, therefore	/img_C08InChp.dd/Homework/NTONY_1.TXT	2007-02-01 10:54:50 EST	0000-00-00 00:00:00

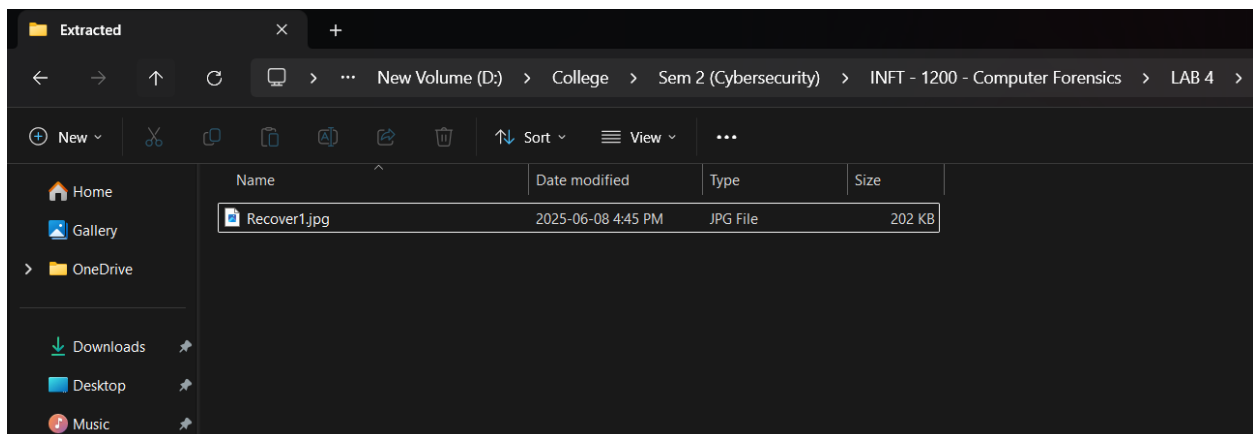
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Metadata

Name: /img_C08InChp.dd/Vacation Pictures/gametour2.exe
 Type: File System
 MIME Type: application/octet-stream
 Size: 206281
 File Name Allocation: Allocated
 Metadata Allocation: Allocated
 Modified: 2001-08-05 07:50:24 EDT
 Accessed: 2017-07-10 00:00:00 EDT
 Created: 2017-07-10 17:55:39 EDT
 Changed: 0000-00-00 00:00:00
 MD5: bb35e45cd178ed16d35e7ca1e1150c00
 SHA-256: ae0de6613b883f50c658eb2a58bee2b4405e23f9c563ac0256034b807c8837b0
 Hash Lookup Results: UNKNOWN
 Internal ID: 212

From The Sleuth Kit istat Tool:

File Extracted:

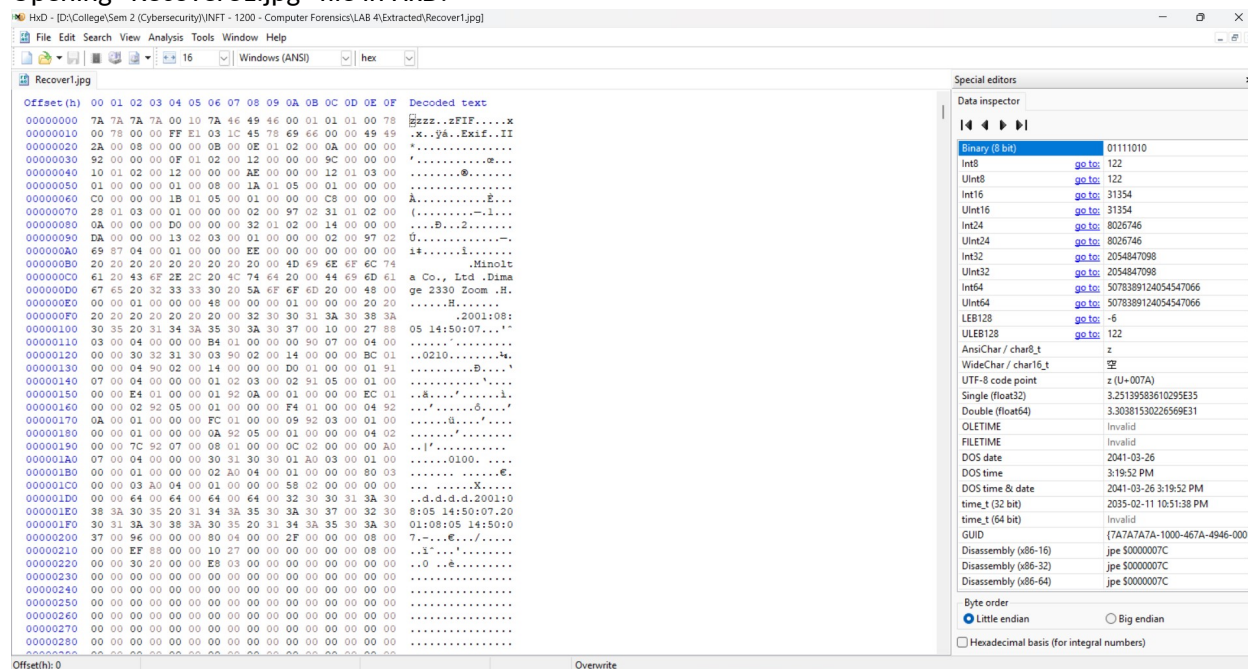


Note: In WinHex Demo, you can save only up to 200 KB of data in a file. However, you can download HxD Hex Editor and recover the above file.

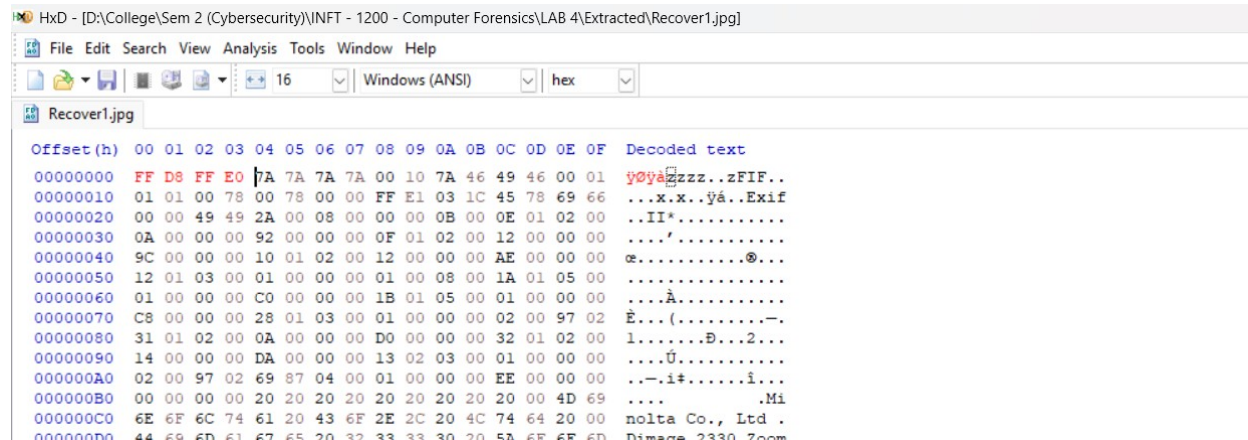
Task 2:

1. Download HxD Hex editor
2. Install HxD Hex Editor
3. Repeat steps 1,2 & 3; in page 357-358

Opening "Recover1.jpg" file in HxD:



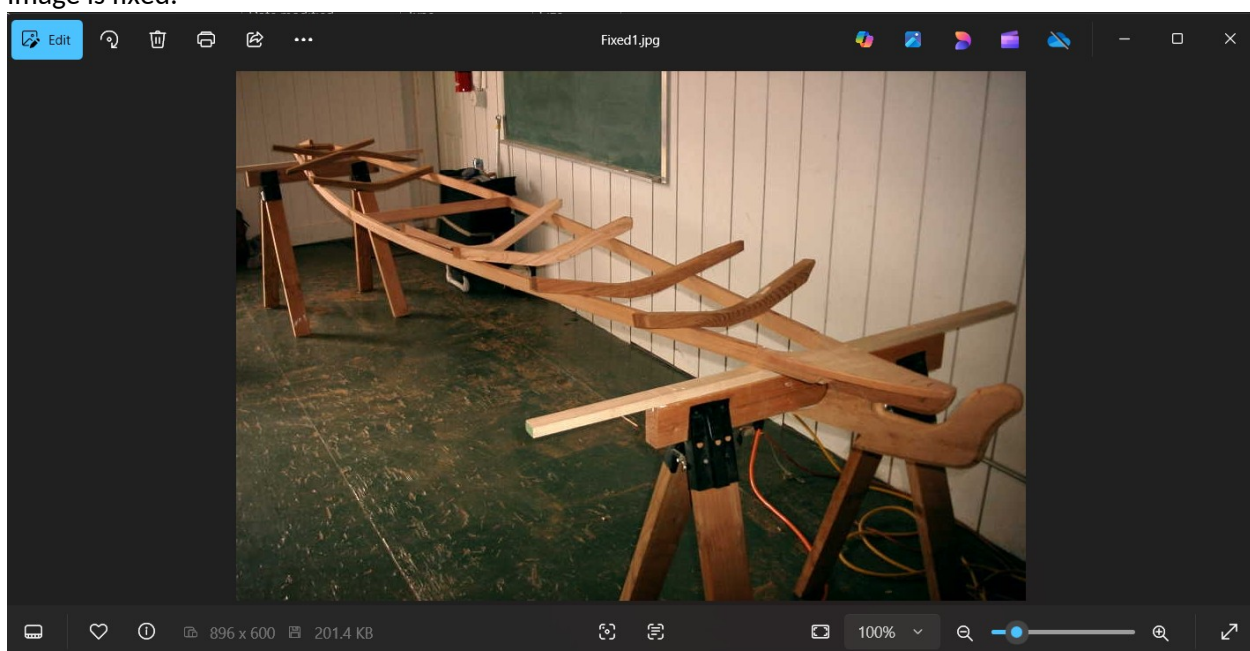
Making changes to the Hex value of file since first 4 bytes at offset 0 are "7A" and usually for a jpeg or jfif file first 4 bytes are "FF D8 FF E0" due to which file was not able to execute as jpg file even though the extension was changed from ".exe" to ".jpg".



Changing "zFIF" to "JFIF":

Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	FF	D8	FF	E0	7A	7A	7A	7A	00	10	4A	46	49	46	00	01	yøÿàzzzz..FIF..
00000010	01	01	00	78	00	78	00	00	FF	E1	03	1C	45	78	69	66	...x.x..ÿá..Exif
00000020	00	00	49	49	2A	00	08	00	00	00	0B	00	0E	01	02	00	..II*.....
00000030	0A	00	00	00	92	00	00	00	0F	01	02	00	12	00	00	00	...'.
00000040	9C	00	00	00	10	01	02	00	12	00	00	00	AE	00	00	00	æ.....@...
00000050	12	01	03	00	01	00	00	00	01	00	08	00	1A	01	05	00	
00000060	01	00	00	00	C0	00	00	00	1B	01	05	00	01	00	00	00	À.....
00000070	C8	00	00	00	28	01	03	00	01	00	00	00	02	00	97	02	È... (.....-
00000080	31	01	02	00	0A	00	00	00	D0	00	00	00	32	01	02	00	1.....ð...2...
00000090	14	00	00	00	DA	00	00	00	13	02	03	00	01	00	00	00	Û.....
000000A0	02	00	97	02	69	87	04	00	01	00	00	00	EE	00	00	00	..-.i#.....î...

Image is fixed:



Reflective statements (end-of-exercise):

You should reflect on these questions:

1. Briefly describe the Exchangeable Image File (EXIF) format.?

Files that have metadata that are automatically embedded into the file are called Exchangeable Image File (EXIF). Examples for metadata are time, date, device information, and if enabled then location information as well as camera settings such as ISO, shutter speed, etc. In digital forensics, EXIF is especially valuable because it can help verify when and where a photo was taken, identify the device

used, and even reveal if the metadata has been tampered with, making it a powerful tool in investigations.

2. Explain how lossless compression relates to image file formats?

By shrinking the file size without losing any of the original information, lossless compression makes it possible to retrieve the full quality of the image upon decompression. When the image quality needs to be retained, for example, in digital forensics, technical drawings, or medical imaging, this kind of compression is critical within image file formats. Lossless compression is commonly used in PNG, TIFF, and occasionally BMP formats. All the image data is stored in these types, so that exact duplication without artefacts is possible. Lossless compression retains all the detail, and for that reason it is suitable for use where accuracy and integrity are necessary. This contrasts with lossy formats like JPEG, which permanently discard some data to save space.

3. Describe how to hide information on an 8-bit bitmap image file using substitution steganography?

In the substitution steganography, the secret message is inserted into an 8-bit bitmap image by replacing the least significant bits (LSBs) of the image pixels with the message bits. As a pixel in an 8-bit bitmap is one byte (8 bits), modifying only the lowest bit (LSB) affects the color of the image very slightly and such modifications are practically invisible to the human eye. So, if the value of one of these pixels is 10101100 and the bit you want to hide is 1 you change that last bit to give 10101101. It is done in the same manner with many pixels, one message bit at a time. Substitution steganography does not change the way the image looks very much, so it is a subtle technique for hiding information in plain sight.

4. Give a brief overview of copyright laws pertaining to graphics within and outside the U.S.?

Copyright law will not allow misappropriation of artwork and other creative works of authorship. Copyright protection exists automatically in the U.S. as soon as a graphic has been fixed and recorded at a substantial level (such as when it's stored, published, posted, etc.). The artist possesses the exclusive right to copy, distribute, display and modify the artwork. Even for educational or personal purposes, infringing copyright on images can have serious legal consequences unless it's protected by an exception, such as fair use.

References

Harvey, P. (n.d.). ExifTool by Phil Harvey. <https://exiftool.org/>

Irvine, B. (n.d.). JPEG image and carving. Asecuritysite. <https://asecuritysite.com/forensics/jpeg>

Wikipedia contributors. (n.d.). Copyright law of the United States. Wikipedia. https://en.wikipedia.org/wiki/Copyright_law_of_the_United_States